

Lab 4 - Newton's second law of motion

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Learning objectives

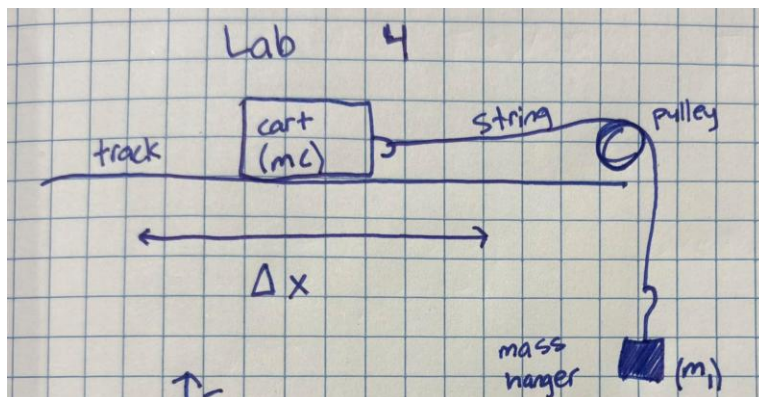
- Observe the relation between the net force and acceleration
- Use spreadsheets to record and analyze data.
- Design and execute an experiment,
- Compare models and experiments.

Equipment

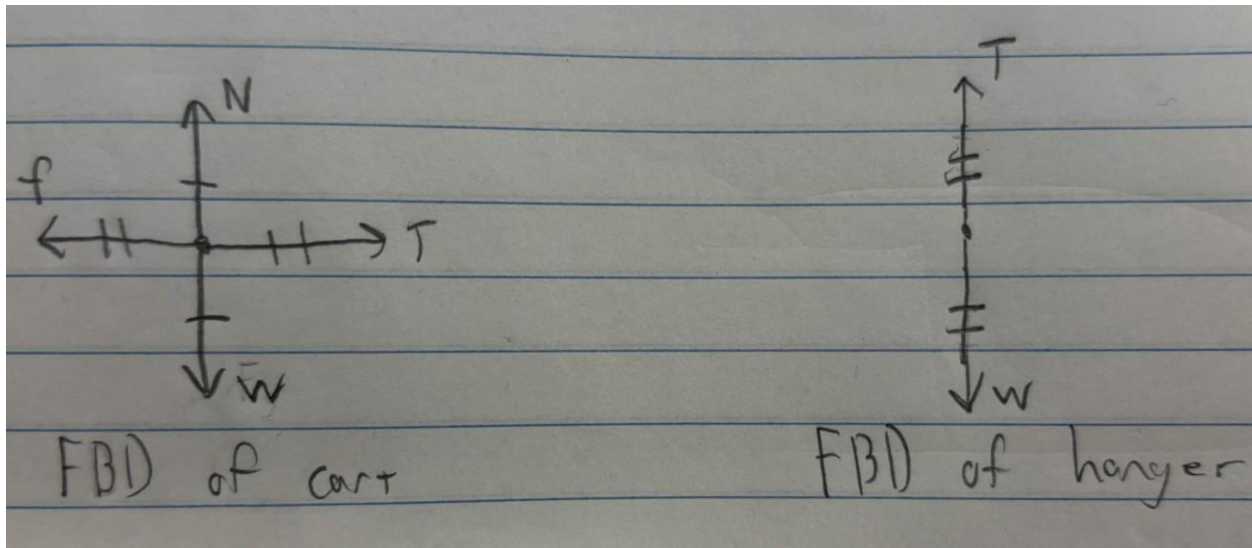
- Capstone data/sensor interface
- Dynamic cart with force sensor
- Dynamic track with bumpers
- Pulley
- String
- Mass hanger 50 g
- Slotted masses 7 x 20 g
- Electronic balance scale

Setup

1. **Set up** the experiment as demonstrated by the instructor.
Label the masses of the cart (**mc**) and the hanger (**m1**).
Attach one end of a length of string to the cart's *force sensor hook on the cart*.
Attach the other end to the mass hanger. **Be sure to** measure and cut enough length of a string such that, when passed over the pulley, **m1** will be able to fall from the pulley all the way to the ground but without too much slack. (Hint: this is a tension experiment!)
2. **Label** the equipment in the diagram shown:



3. **Draw** two Free Body Diagrams, one for the cart and another for the mass hanger when $a = 0$. Insert screenshots.



4. **Measure** and **record** m_c on the electronic balance scale in SI units with 4 sigfigs. **Measure** and **record** m_1 on the electronic balance scale. Check that the measured mass matches the marking. This is a good check for how **accurate** the scale is.

$$m_c = 0.2497\text{kg}$$

$$m_1 = 0.0492\text{kg}$$

Prelab

We will do the prelab as a class to get familiar with how to **zero the force sensor** and take measurements.

1. **Setup** the Capstone interface **as demonstrated by the instructor** with two plot areas on one graph display. **Select** time (s) for the x-axis of the graph. Select velocity (m/s) for the top y-axis and Force (N) for the bottom y-axis. **Select** "Smart Cart Force Sensor" at the bottom and the frequency to 20.00 Hz.
2. **Think about and answer as a group**
Which force in the FBDs does the force sensor measure?
Tension
When should the force sensor read 0 N?
When there is no tension on the cart

3. **Each person in the group** practice zeroing out the force sensor.
(you must click the “ 0 ” button under the right condition.

Experiment 1

Hold the cart stationary and start recording on Capstone. Then release the cart from rest. Stop the recording before the cart hits the bumper.

Your velocity versus time plot should look flat at first and constantly increasing after the time of the release. Your force versus time graph should have two flat sections, with the first section having a larger magnitude compared to the section after the time of the release.

RECORD DATA IN A GOOGLESHEET

1. Record the weight of m_1 , aka $m_1 \cdot g$. READ OFF THE GRAPH! This should be the average magnitude of the force on the cart before the release. Highlight the range of points in the graph to find the average (mean) . DO NOT MULTIPLY THE MASS OF THE HANGING MASS BY 9.81!! READ THE VALUE OFF THE GRAPH OF FORCE VS TIME.

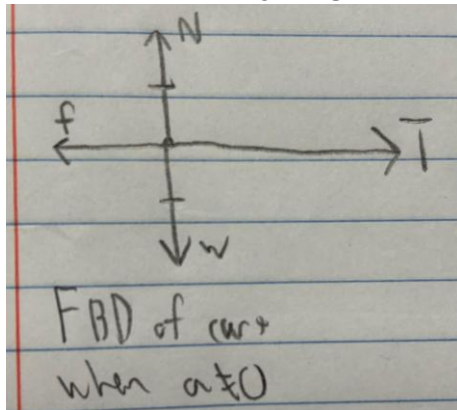
2. Record the force on the cart after the release, aka F_1). Highlight a new range to find this average (it should be different). If not, ask for help.

3. Record the acceleration of the cart, aka slope of the v-t curve after the time of release while the cart is accelerating. a should not be zero. If it is, ask for help

4. Increase the mass of m_1 , repeat the experiment at least 7 times (with a different mass of m_1 each time). Each time, record $m_1 \cdot g$, F_1 and a in the spreadsheet.

Analysis

1. Draw a free-body diagram of the cart during acceleration.



2. Plot F_1 v. a in your googlesheet.

What kind of curve best describes this relationship?

The relationship appears to be linear.

Is this what you expect from Newton's second law of motion? Explain.

This is what we expect since, according to Newton's second law of motion, as the mass of an object increases, its force increases proportionally.

What is the equation of the best-fit straight line? (select "show equation" of trendline).

LINEAR = $y = mx + b$

$m = \text{slope}$ $b = y\text{-int.}$

$Y = 0.254x + 0.0109$

What physical quantity is represented by the slope? (Hint: look at the units or remember Newton's second law)

The physical quantity is the mass of the cart.

Compare the value of the slope to the actual value of the physical quantity it represents.

The mass of the car does match the measured mass of the car, which is 0.254 kg

Based on the best-fit straight line, what is the value of the acceleration when $F_1 = 0$?

If the acceleration is non-zero when $F_1 = 0$, what is this evidence of?

$0.254x + 0.0109$

$= 0.254(0) + 0.0109$

$= 0.0109$

A non zero acceleration is evidence of friction acting on the cart

Based on the best-fit straight line, what is the value of the F_1 when $a = 0$? If it is non-zero, why not?

F_1 when $a = 0$ is 0 according to our line of best fit (with uncertainty)

Based on your results, is friction negligible in this experiment? If not, what is your best estimate for the value of the friction force?

No, the friction is not negligible; our best estimate for the force of friction acting on the cart is -0.008 N

$F_1 - F_f = ma = F_{\text{net}}$

$0.50\text{N} - F_f = (0.254\text{ kg})(2.000\text{ m/s}^2)$

$-F_f = 0.008\text{ N}$

$F_f = -0.008\text{ N}$

Revise your FBD if needed. Based on this FBD, what is the net force acting on the cart? (Express your answer in terms of F_1 and F_f , the friction force)

$F_{\text{net}} = F_1 - F_f$

F_{net} is equal to the force of the mass hanger descending (F_1) subtracted by the force of friction opposing (F_f)

Apply Newton's second law to write the cart's equation of motion in terms of m_c , F_1 , F_f , and a .

$$a = \frac{1}{m_{\text{cart}}} (F_1 - F_f)$$

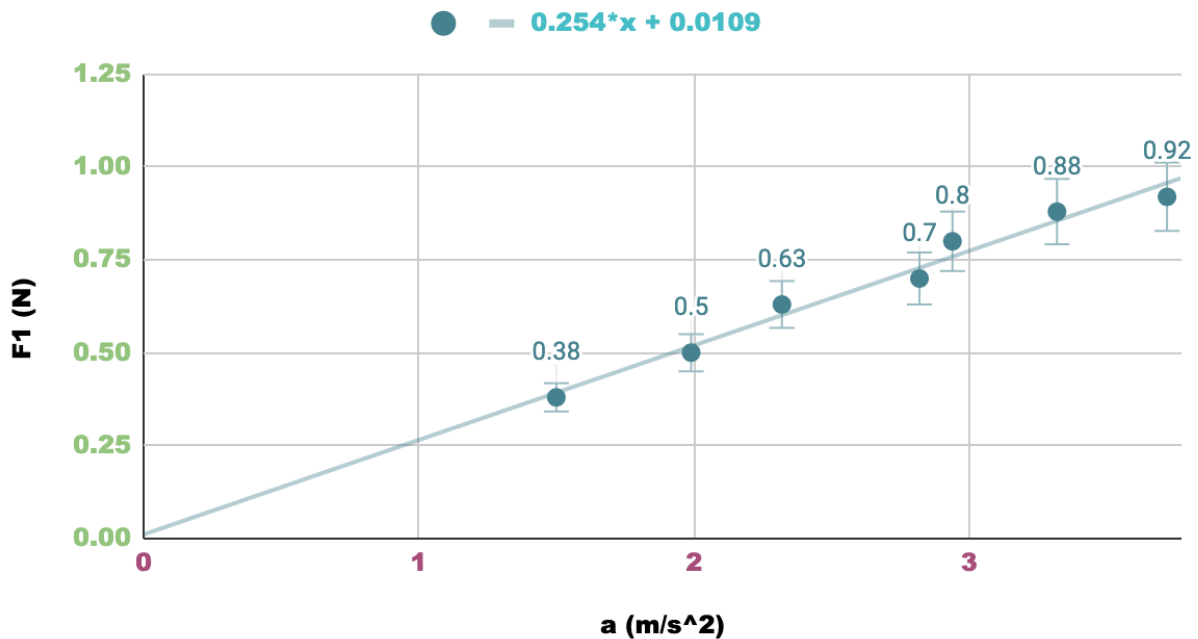
Since both $m_c(a)$ and $F_1 - F_f$ are both equal to F_{net} we can set them equal to each other and then solve for acceleration to find the cart's motion

Is F_1 equal to $m_1 \cdot g$? If not, explain. (Hint: draw a FBD of the hanging mass).

F_1 is not equal to $m_1 \cdot g$ because for the mass hanger to descend pulling the cart its weight has to be heavier than the tension on the line.

Experiment #1					
Run #	Mass m_1 (kg)	Weight of m_1 (N)	F_1 (N)	Acceleration of cart (m/s^2)	
	m_1 (kg) (mean)	$m_1 \cdot g$ (N)	F_1 (N)	a (m/s^2) (linear fit)	
1	0.0494 kg	0.48	0.38	1.5	
2	0.0673 kg	0.66	0.5	1.99	
3	0.0893 kg	0.86	0.63	2.32	
4	0.1092 kg	1.05	0.7	2.82	
5	0.1292 kg	1.25	0.8	2.94	
6	0.1491 kg	1.44	0.88	3.32	
7	0.1691 kg	1.63	0.92	3.72	

F1 (N) vs a m/s²)



Experiment 2

Design and execute an experiment that shows that the acceleration is inversely proportional to its mass. Hint: simplify the data acquisition process from Experiment 1 to measure only the variables of interest. (only need to record acceleration, not force. Then plot a vs. m ; m is mass of the cart)

Design

The cart will be on the track with a mass hanger hanging off the edge to create tension. Weight will then be added to the cart each trial.

Use the data generated by your experiment to plot a vs m .

No data collected (not enough time)

What do you expect the relation between a and m to be?

As the mass of the cart increases, the acceleration should decrease. This is because Newton's second law, solved for acceleration, shows them as inversely proportional.

Fit a power series to your plot. Does this fit match your expectation?

No data collected (not enough time)

Can you explain any difference? (Hint: Was $F_{\text{net}} = mc \cdot a$ actually constant? what other unaccounted variable may have changed as you were increasing m ?)

No data collected (not enough time)lalo

Experiement #1

Run #	Mass m1 (kg)	Weight of m1 (N)	F1 (N)	Acceleration of cart (m/s^2)
	m1 (kg) (mean)	m1*g (N)	F1 (N)	a (m/s^2) (linear fit)
1	0.0494kg	0.48	0.38	1.5
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